Field Determination of Resilient Pressure Stiffness (k_p)

Condensed operating procedure and geometric reference sheet

This briefing turns the original note into a print-first field handout. Key equations are isolated into bordered math cards and the procedure is condensed into an operational checklist.

- Plate Diameter: 0.3 to 0.45 m
- Target Pressure: 1g tire bearing pressure
- Typical Cycling Rate: 0.5 to 1.0 Hz

FIELD PROCEDURE

Execution sequence

01 Prepare and place the plate

Select representative haul-road locations, smooth the immediate contact zone, and seat the bearing plate squarely on the running surface.

02 Match the target tire pressure

Set the cyclic loading target to the nominal 1g tire bearing pressure for the haul truck being modeled.

03 Cycle until deformation converges

Repeat unload-reload cycles until the resilient portion of the deformation stabilizes and the response becomes repeatable.

04 Record the resilient rebound response

Use the peak cyclic stress and the resilient elastic displacement from the converged cycle to compute k_p.

EQUIPMENT

Required field setup

- Cyclic plate load assembly with known footprint area.
- Hydraulic ram with reliable pressure readout.
- Independent ground-referenced displacement measurement.
- Data acquisition system to confirm cyclic convergence.

PRIMARY STIFFNESS EQUATION

Resilient pressure stiffness

$$k_p = \frac{\Delta\sigma}{\Delta\delta_{\text{resilient}}}$$

Use the peak cyclic pressure and the resilient elastic deformation measured from the converged cycle.

FOOTPRINT AREA

Empirical tire contact relation

$$A = 1.35 \times \delta_{tire} \times \phi$$

The area is inferred from tire deflection and diameter rather than measured directly in the field.

FOOTPRINT CHORD

Contact patch length

$$L = 2\sqrt{(\frac{\phi}{2})^2 - (\frac{\phi}{2} - \delta_{ti}}$$

This is the cleanest print form of the chord equation because it exposes the underlying circle geometry directly.

INTERPRETATION NOTES

How the geometry feeds the rolling resistance model

- The simulator only needs a few primary drivers: tire load, tire stiffness, tire diameter, and resilient ground stiffness.
- From those values it can derive tire deflection, footprint area, footprint chord length, and the geometric interaction arc.
- Separating the equations into isolated math cards avoids the dense inline notation that becomes difficult to read in PDFs.

Prepared as a compact field reference for the haul-road rolling resistance workflow.